MILITARY SPECIFICATION

CLOTH, TWILL, NYLON WARP. RAYON FILLING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a twill nylon, rayon fabric.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date on invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-P-1133	Packaging and Packing of Synthetic Fiber Fabrics.

STANDARDS

Federal

FED-STD-4	Glossary of Fabric Imperfections
FED-STD-191	Textile Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.3 and 6.5).

3.3 Material

3.3.1 Yarn. The warp shall be 200 denier, trilobal nylon. The filling yarn shall be 14/2s spun viscose rayon.

3.4 Color. The color shall be Air Force Blue 1157.

3.4.1 Matching. The color of the finished cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.4.2 Colorfastness. The finished cloth shall show fastness to wet dry cleaning, crocking, perspiration, and light equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished cloth shall show "good" fastness to wet dry cleaning, perspiration, and light and shall show a Munsell Value for crocking not less than 8.5.

3.5 Physical requirements. The finished cloth shall conform to the requirements specified in Table I when tested as specified in 4.5.

TABLE I PHYSICAL REQUIREMENTS

Weight ozs./sq. yd. (minimum)	Yarns per inch (minimum)		Breaking strength (minimum)		Shrinkage percent (maximum)		Tearing strength (minimum)	
	Warp	Filling	Warp	Filling	Warp	Filling	Warp	Filling
8.3	138	44	340	290	1.5	1.0	25	20

3.5.1 Width. The width of the finished cloth shall be a minimum of 45 inches exclusive of selvage.

3.5.2 Weave. The weave shall be a 3 up and 1 down, 4 harness left twill.

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3.6 Finish. The fabric shall be given a durable, flourocarbon type anti-soiling treatment.

3.7 Nonfibrous materials. The total chloroform-soluble and water-soluble material content of the finished cloth shall not exceed 1.0 percent when tested as specified in 4.3.

3.8 pH. The pH value of the finished cloth shall be no lower than 5.0 nor higher than 8.5 when tested as specified in 4.4.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 50 yards. Each length shall be put up on full width rolls as specified in PPP-P-1133.

3.10 Marking. The back of the cloth shall be marked "Back" at both ends of each roll with letters not less than ½ inch in height (see 4.4.3.3). The marking shall be clearly legible in any indelible marking used commercially.

3.11 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards, when calculated as specified in Section 4, shall not exceed the established maximum point value. Each defect shall be marked with a 1 to 1½ inch long string. The string shall be sewn into the selvage near the defect. A red string shall represent a three or four point defect and a white string shall represent a one or two point defect. (see 4.4.1.1.1 and 4.4.1.5).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When required (see 6.2), the preproduction sample submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish, and shall be tested in accordance with the methods specified in 4.5.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.4.1 Examination of the end item. Examination of the end item shall be in accordance with 4.4.1.1 through 4.4.1.5.

4.4.1.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section III of Fed-std-4, which are clearly noticeable at normal inspection distance (3feet) shall be scored and assigned demerit points as listed in 4.4.1.1.1. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than 4 points. The sample size shall be in accordance with the following:

Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample.

The lot shall be unacceptable if the points per 100 square yards of the

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total yardage examined exceeds 10.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 15.0 points. If one roll exceeds 15.0 points per 100 square yards, a second sample shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 15.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

4.4.1.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Bowed or bias filling exceeding 2 inch deviation from normal alignment

Baggy, ridgy or wavy cloth

Objectionable odor

Width less than specified

Uneven weaving

Tight selvage

Tight warp end

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4.4.1.2 Examination for length.

4.4.1.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective with respect to length.

4.4.1.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.4.1.3 Examination for face identification and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each roll in the sample shall be examined for the defects listed below. The lot shall be unacceptable if two or more of the following defects are present:

Face identification missing from either or both ends

Face identification on wrong side

Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

4.4.1.4 Examination for shade and finish

4.4.1.4.1 Individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.4.1) and finish (3.6). A roll shall be unacceptable if it fails to meet the requirements for shade match and finish. The sample unit shall be a 4-inch by 20-inch swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

4.4.1.5 Examination for defect marking. During the yard-by-yard examination, each roll in the sample shall be examined for defect marking. Any roll having more than 2 three or four point fabric defects (red string) not strung or more than 5 fabric defects red or white strings

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not strung shall be classified as a defective roll in regard to the defect marking requirement (see 3.9). The lot shall be unacceptable if more than two rolls in the sample are defective in regard to the defect marking requirement or if there are more than 10 three or four point defects (red strings) not strung in the sample.

4.4.2 Examination of preparation for delivery requirements. Examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing, and marking comply with the section 5 requirements.

4.5 Tests. The methods of testing specified in Fed-Std-191 wherever applicable, and as listed in Table II shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 3½ continuous yards full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size (Yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II Test Methods

Characteristic	Requirement paragraph	Test method
Yarn :		
Nylon (trilobal) identification	3.3.1	1/
Denier	3.3.1	1/
Ply (filling)	3.3.1	1/
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050

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Table II Test Methods (continued)

Characteristic	Requirement paragraph	Test method
Weight	3.5	5041
Breaking strength		
Warp	3.5	5100
Filling	3.5	5100
Tearing strength		
Warp	3.5	5134
Filling	3.5	5134
Weave	3.5.2	Visual
Finish	3.6	2/
Shrinkage	3.5	5558

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ The contractor shall identify the type of finish used.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

5.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order shipments shall be marked in accordance with PPP-P-1133.

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   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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